

GLOSSARY

Terms defined elsewhere in the glossary are in *italics*.

aberration Any of various defects that may occur in an image formed by a lens or mirror.

absolute scale Also called Kelvin scale, a temperature scale starting at *absolute zero*.

absolute zero The lowest possible temperature [0 K, -459.67°F , or -273.15°C], when there is no random energy of movement in atoms and molecules.

absorption (1) The taking up of one substance by another. (2) The capture of electromagnetic radiation by matter.

acceleration The rate of change of velocity.

acid A compound containing hydrogen that splits up in water to give reactive hydrogen ions.

acoustics (1) The study of sound. (2) The properties of a particular space, such as a concert hall, in terms of how sound travels around it.

active transport In biology, any transport of substances across cell membranes that requires energy input.

acupuncture A medical treatment originating in China, which involves fine needles being inserted into the skin at particular points.

adaptation Any inherited aspect of an organism's structure or behavior that helps fit it to its environment; also, the evolutionary process giving rise to such features.

ADP Adenosine diphosphate, a compound formed when *ATP* releases energy.

adrenal gland Either of two glands situated one on top of each kidney.

alchemy A medieval science that tried, among other things, to change different metals into gold.

algae [singular: **alga**] Simple water-living organisms that produce their food by photosynthesis. They include single-celled forms as well as large seaweeds.

algebra A branch of mathematics that involves performing calculations using letters and other general symbols.

algorithm Set of rules by which calculations can be performed automatically, especially by a machine, to get a particular result.

alkali A *base* that dissolves in water.

alkaline A solution with a pH greater than 7.

allotropes Different forms of the same element. For example, graphite and diamond are allotropes of carbon.

alloy A metal that is a mixture of more than one element, either all metallic elements or with non-metals mixed in.

alternating current An electric current whose direction reverses at regular intervals.

alternator An electric generator that produces an alternating current.

amino acid Any of a group of small molecules that are the building blocks of proteins. They also have various other roles in the body.

amniocentesis Obtaining a sample of the fluid surrounding a baby in the womb by

passing a hollow needle through the mother's body wall under anesthetic.

amp (ampere) The SI unit of electric current.

amplitude The size of a vibration or the height of a wave.

anaphase The stage of *mitosis* or *meiosis* where *chromosomes* or chromatids separate from one another.

anatomy The study of the internal structure of living things.

anesthesia The production of pain relief by loss of sensation or consciousness. Drugs that achieve this are called anesthetics.

angle of incidence The angle between a light ray hitting a surface and an imaginary line perpendicular to that surface.

angle of reflection The angle between a light ray reflected from a surface and an imaginary line perpendicular to that surface.

anion A negatively charged *ion*.

anode A positive *electrode*.

anther A pollen-producing structure of a flower, which together with its supporting stalk (filament) makes up a stamen.

antibiotic A drug used to kill or inhibit the growth of bacteria that cause infections.

antibodies Proteins produced by the body that identify and attack foreign particles, such as invading bacteria.

antigen Anything that stimulates the body to produce antibodies, such as the outer coat of an invading microorganism.

antiparticle A version of a subatomic particle that is opposite in electrical charge to the normal version.

aphasia The inability to produce and/or understand speech.

area The size of a two-dimensional surface.

arithmetical progression A sequence of numbers each differing from the previous one by the same amount.

armillary sphere A metal model in the form of an open sphere, representing the apparent movements of the Sun, stars, planets, etc. as seen from Earth.

artery A blood vessel leading away from the heart. See also *vascular circulation*.

asteroid A small rocky body orbiting the Sun. See also *meteoroid*.

asthenosphere The relatively soft upper layer of Earth's mantle, below the *lithosphere*.

astrolabe A historical astronomical instrument used by astronomers and sailors for locating the Sun, Moon, planets, and stars.

astronomical unit A unit of distance used in astronomy equal to the distance between Earth and the Moon.

astronomy The scientific study of space and the Universe beyond Earth's atmosphere.

atmosphere (1) The gases that surround the Sun, Earth, and some planets. (2) A measurement of pressure.

atmospheric pressure The normal pressure of the air, especially near the ground.

atom The smallest part of an element that has the chemical properties of that element. An atom consists of a nucleus of protons and neutrons surrounded by orbiting electrons.

atomic mass Also called atomic weight, a measure of the relative amount of material in different kinds of atoms. (Hydrogen atoms have the smallest atomic mass.)

atomic number The number of protons in the nucleus of an *atom*. All atoms of the same element have the same atomic number.

atomic theory Any theory that states that matter is made up of atoms.

atomic weight See *atomic mass*.

ATP Adenosine triphosphate, an important energy-carrying *molecule* in all living cells.

aurora borealis A display of lights in the night sky in Arctic regions, caused by the impact of electrically charged particles from the Sun with Earth's atmosphere.

axis (1) The imaginary line about which a body, such as a planet, rotates. (2) A reference line on a graph.

axle A structural rod upon which a wheel or wheels revolve, or a rod which revolves along with the wheels.

background radiation See *cosmic background radiation*.

bacteriophage A virus that attacks bacteria, called phage for short.

bacterium [plural: **bacteria**] A member of a kingdom of single-celled microscopic life forms whose cells lack nuclei. See also *prokaryotic cell*.

barometer An instrument for measuring air pressure.

base (1) A substance that reacts with an *acid* to form a salt. (Soluble bases are called alkalis.) (2) Any of the four similar molecules found repeatedly in molecules of DNA, whose order "spells out" the genes of living things. (3) In mathematics, the specific number that forms the basis of how numbers are conventionally written down. For example, in everyday life 10 means "ten" (= one ten and no units) but in the binary system 10 means two (= one two and no units); this latter system is also called "base 2".

base pair A matching pair of complementary bases opposite each other on the strands of a DNA molecule.

battery Originally, two or more voltaic cells connected together; now often just means a single voltaic cell.

beta decay A form of radioactive decay in which beta particles (fast-moving electrons or positrons) are given out.

Big Bang The moment, estimated at about 13.8 billion years ago, when the present Universe is thought to have begun by explosion and expansion from a tiny point.

binomial system The standard system of giving a two-part scientific name to each biological species. For example, the human species is named *Homo sapiens*.

biodegradable Able to be broken down by natural biological processes.

biomass (1) The amount of living material of a specific kind or in a given area. (2) Non-fossil plant material, such as wood, usable as fuel.

biosphere The surface regions of Earth where living things are found.

bit In computing, a fundamental unit of information having just two possible values, as either of the binary digits 0 or 1.

black body A theoretical object that can absorb all electromagnetic radiation, and can also emit radiation of all wavelengths, dependent only on its temperature.

black hole A super-dense body of matter with gravity so intense that not even light can escape from it.

blastocyst A hollow ball of cells that is an early stage in the formation of an embryo.

blood type Also called blood group, any of several categories into which blood can be classified, defined by differences in the surface chemistry of red blood cells.

blood vessel An *artery*, *vein*, or *capillary*. See also *vascular circulation*.

boiling point The temperature at which a particular liquid changes into a gas.

bond A binding connection between atoms.

botany The study of plants.

Brownian motion The random movement of tiny particles in a liquid or a gas, caused by molecules colliding with them.

buoyancy The tendency of an object to rise upward in a fluid (liquid or gas) when the object is less dense than the fluid.

byte A unit of information storage and transmission in computing and telecommunications. A kilobyte is a thousand bytes, a megabyte is a million bytes, and a gigabyte is a billion bytes.

calculus (1) A branch of mathematics based on calculations involving tiny infinitesimal changes. It comprises differential calculus, which is concerned with rates of change, and integral calculus, which can be used to calculate areas, volumes, etc. (2) A medical name for a hard mass formed in the body, such as a kidney stone.

calendar round cycle In the Maya civilization, a cycle of 52 years after which the two separate Mayan calendar systems become aligned with one other.

calx A powdery or crumbly substance left when a mineral or metal has been burnt.

capacitor A device used to store electric charge temporarily.

capillaries The tiny blood vessels that supply tissues and connect the arteries and veins. See also *vascular circulation*.

carbon A chemical element (symbol C, atomic number 6) which forms more compounds than any other element, including the important chemicals of life.

carbon cycle The cycling of carbon through the living and non-living parts of Earth and its atmosphere.

cartography The science and practice of mapmaking.